Day 2: Tasks

1. **List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.**

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Device	Type Of Power Supply	Reason
Desktop Computer	SMPS	It converts AC power into the DC voltages required by computer components.
Leptop Charger	SMPS	It provides efficient and stable power while charging the laptop.
LED TV	SMPS	It supplies regulated power to the TV's internal circuits.
Desktop Computer	UPS	It keeps the computer on during a power cut.
Wi-Fi Router	UPS	It keeps the internet working when the power goes out.

2. **Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.**

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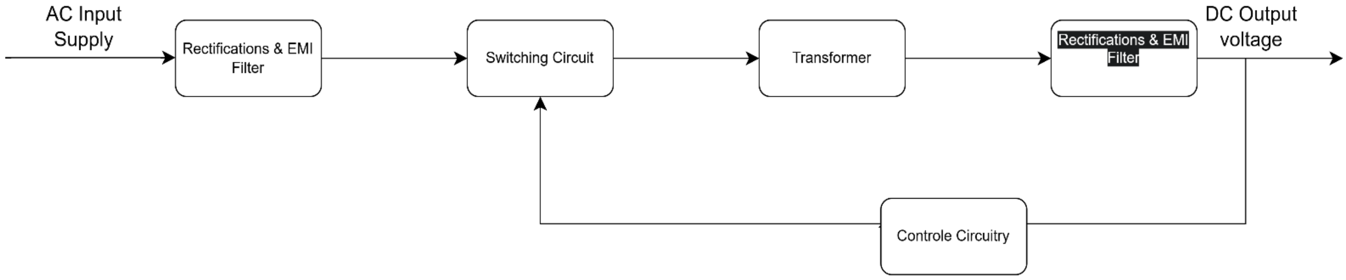

3. **Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.**

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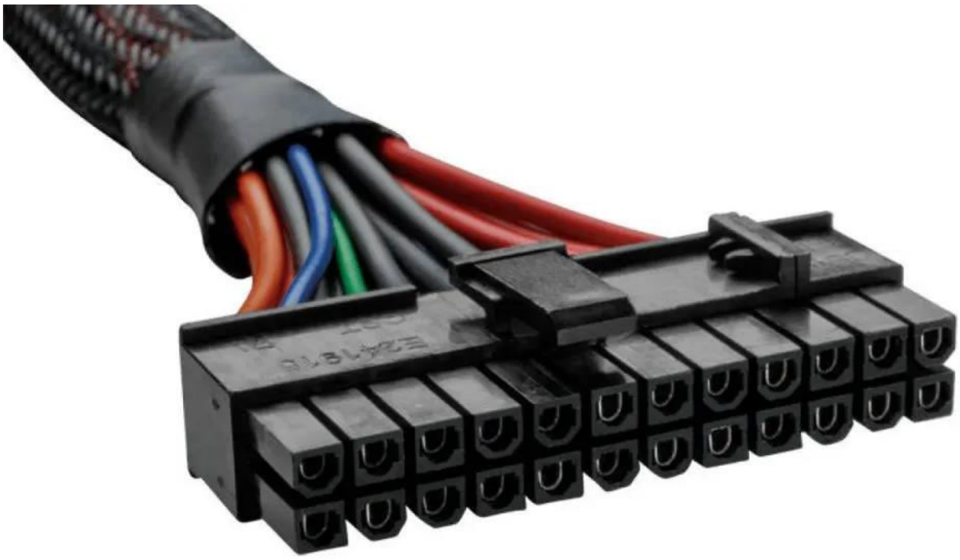
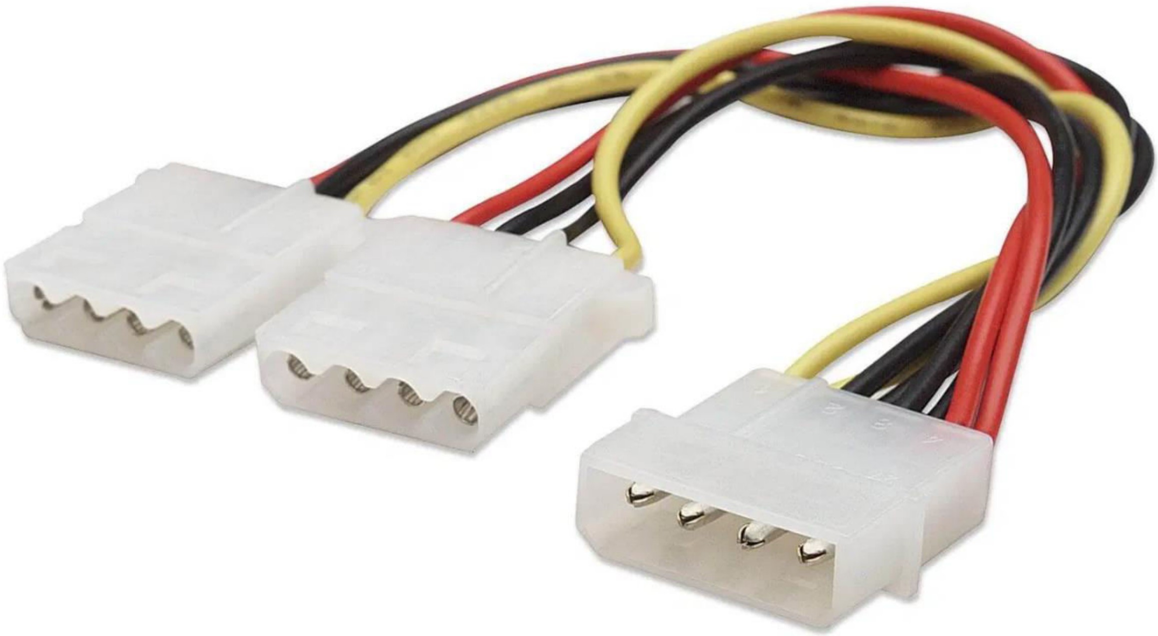
- **Connector Name : SATA Connector *(SATA Data Cable)***
- **Devices it powers : Connects SSDs and HDDs to the motherboard for data transfer.**
- **safety tip : Do not bend the cable sharply, as it may damage the connection.**



- **Connector Name : ATX 24-pin Motherboard Connector**
- **Devices it powers : Provides main power to the motherboard.**
- **safety tip : Always switch off and unplug the power supply before connecting or disconnecting it.**

- **Connector Name : PCIe 6/8-pin Connector**
- **Devices it powers : Powers a graphics card (GPU)**
- **safety tip : Make sure the connector is fully inserted before turning on the PC.**



- **Connector Name : Molex Connector**
- **Devices it powers : Powers older hard drives, optical drives, and cooling fans.**
- **safety tip : Pull the connector gently by the plug, not by the wires..**

4. **Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.**

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Steps to Connect the SMPS in a New Gaming PC :

1. **Install the SMPS** inside the computer cabinet and secure it with screws.
2. **Connect the Motherboard Power**
   - Use the **24-pin ATX power cable** from the SMPS
   - Plug it into the **24-pin ATX connector** on the motherboard
3. **Connect CPU Power**
   - Use the **8-pin EPS cable** from the SMPS.
   - Connect it to the **CPU power connector** near the processor socket on the motherboard.

4. **Connect the Graphics Card**
    - Install the graphics card in the PCIe slot.
    - Use the **6-pin or 8-pin PCIe power cable**
    - Connect it to the power connector on the graphics card.
5. **Connect Storage Devices**
    - For SSDs or HDDs, use the **SATA power cable** from the SMPS.
    - Connect it to the **SATA power port** on the storage device.
    - Also connect a **SATA data cable** from the storage device to the motherboard.
6. **Check All Connections**
    - Ensure all cables are firmly connected.
    - Organize cables properly for good airflow inside the cabinet.
7. **Power On the PC**
    - Connect the power cord to the SMPS and switch it on.
    - Start the PC and verify that all components are working correctly.